From Optimal Transport to Trajectory Inference

Entropic transport, Schrödinger bridges and mean-field Langevin dynamics: a handbook for readers who know OT

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Abstract

Trajectory inference (TI) asks how individuals in a population move over time when the population can only be observed through *snapshots*, as in single-cell RNA sequencing, where measuring a cell destroys it. Mathematically, TI is a problem of selecting a law on path space from a sequence of its marginals, and optimal transport (OT) enters at exactly that point: OT is a principled way of choosing couplings. This handbook develops the theory of TI from the perspective of a reader who already knows OT and Wasserstein gradient flows. We go from Kantorovich's problem to entropic OT and Sinkhorn's algorithm, from there to the static and dynamic Schrödinger problems, and arrive at the min-entropy estimator of Lavenant et al., a multi-marginal Schrödinger problem that is consistent for drift-diffusion processes. We then explain how Chizat et al. compute it with mean-field Langevin dynamics, a Wasserstein gradient flow on a product of probability spaces whose global convergence rests on *flat* rather than *displacement* convexity, and how unbalanced transport (Wasserstein–Fisher–Rao / Hellinger–Kantorovich) accounts for cell division and death. Throughout, "Connection" boxes make explicit the bridges between these topics, and the last section collects open problems where the OT and gradient-flow viewpoints are likely to be decisive.

Contents

1 Why trajectory inference is a transport problem 3
1.1 The data: snapshots of a population 3
1.2 The missing information is a coupling 3
1.3 A first answer: Waddington-OT 3
2 Optimal transport essentials 4
2.1 Static OT and duality 4
2.2 Dynamic OT and the geometry of $\mathcal{P}_2$ 4
3 Entropic OT and Sinkhorn's algorithm 5
3.1 Definition, dual and the Schrödinger system 5
3.2 Three properties that matter for TI 6
4 The Schrödinger problem 7
4.1 Schrödinger's thought experiment 7
4.2 From dynamic to static: the bridge decomposition 7
4.3 Stochastic control and the fluid picture 7
4.4 Dictionary 9
5 Trajectory inference as a multi-marginal Schrödinger problem 9
5.1 The population model and why it is a gradient flow 9
5.2 The min-entropy estimator 9
5.3 The representer theorem: from paths to T marginals 10
6 Two gradient flows: mean-field Langevin dynamics 11
6.1 Two clocks 11
6.2 Mean-field Langevin dynamics 11
6.3 Why it converges: flat convexity meets Wasserstein dissipation 12
7 Mass is not conserved: unbalanced transport and growth 13
7.1 Growth in the population model 13
7.2 Unbalanced optimal transport 13
7.3 The gradient-flow view of growth 14
8 The wider landscape 14
9 Open problems for the OT and gradient-flow community 15
References 15

1 Why trajectory inference is a transport problem

1.1 The data: snapshots of a population

Let $\mathcal{X} \subset \mathbb{R}^d$ be a state space (in biology: gene-expression space, typically after dimensionality reduction) and $\Omega = C([0,1], \mathcal{X})$ the space of continuous paths. A population of particles evolves according to an unknown stochastic process $(X_t)_{t\in[0,1]}$ with law $P \in \mathcal{P}(\Omega)$. Writing $e_t : \omega \mapsto \omega(t)$ for the evaluation map, the time marginals are $P_t = (e_t)_{\#}P$ and the two-time couplings are $P_{s,t} = (e_s, e_t)_{\#}P$.

Definition 1 (Trajectory inference). Given times $0 \le t_1 < \cdots < t_T \le 1$ and empirical measures

$$\hat{\mu}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \delta_{X_{t_i,j}}, \quad i = 1, ..., T, \tag{1}$$

where all samples are independent (**each particle is observed at most once**), estimate the law $P \in \mathcal{P}(\Omega)$.

The final clause is what separates TI from classical statistics of stochastic processes, where one observes whole (discretised) trajectories. In single-cell RNA sequencing the measurement is destructive, so the data carry information about marginals only (Schiebinger et al., 2019; Weinreb et al., 2018).

1.2 The missing information is a coupling

Marginals never determine the law of a process. Both of the following have $P_t = \mathcal{N}(0,1)$ for every t: (a) the static process $X_t \equiv X_0 \sim \mathcal{N}(0,1)$; (b) the stationary Ornstein–Uhlenbeck process $\mathrm{d}X_t = -X_t\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$. Their couplings $P_{s,t}$ are, respectively, the identity coupling and a Gaussian coupling with correlation $e^{-|t-s|}$. More generally, for a Markov process the law is determined by its marginals *together with* its transition kernels, so the unknown in TI is exactly the sequence of couplings $\left(P_{t_i,t_{i+1}}\right)_i$, or rather a choice among the many couplings compatible with the data.

Connection 2 (TI is coupling selection). Choosing a coupling $\pi \in \Pi(\mu,\nu)$ between two given marginals by a variational principle is precisely what optimal transport does. TI therefore "is" an OT problem as soon as one fixes a principle for picking couplings. The whole theory below can be read as a sequence of increasingly faithful answers to the question *which* variational principle is the right one for noisy, branching populations, and *how* to compute its solution.

1.3 A first answer: Waddington-OT

The first OT-based method, Waddington-OT (Schiebinger et al., 2019), couples each pair of consecutive snapshots by (unbalanced, entropically regularised) OT, $\gamma_i \in \arg\min_{\gamma\in\Pi(\hat{\mu}_i,\hat{\mu}_{i+1})} \int c\,\mathrm{d}\gamma + \varepsilon H(\gamma)$, and composes the resulting transition kernels into trajectories. It was strikingly successful on stem-cell reprogramming data. However, it was a heuristic: why entropic regularisation, why this cost, how to choose ε, and what is estimated when samples are scarce?

Lavenant et al. (2021) answered these questions by identifying the right object: the **min-entropy estimator**, a law on paths minimising relative entropy with respect to Brownian motion while fitting the snapshots. It is consistent for drift-diffusion processes, and it reduces to Waddington-OT in a limiting regime (Section 5). Chizat et al. (2022) then showed how to compute it with a grid-free, provably convergent particle method (Section 6). The route there passes through a chain of classical ideas from OT:

Step	Mathematical object	Role in TI
Section 2	Kantorovich OT, Benamou–Brenier, Otto calculus	zero-noise coupling principle; geometry for gradient flows
Section 3	entropic OT, Sinkhorn, Schrödinger potentials	noise-aware couplings; algorithms; first variations
Section 4	static and dynamic Schrödinger problem	the "most likely" evolution of a Brownian cloud
Section 5	multi-marginal Schrödinger problem with soft constraints	the min-entropy estimator
Section 6	mean-field Langevin dynamics	a convergent gradient-flow solver
Section 7	unbalanced OT (WFR / HK)	cell division and death

Table 1: Roadmap of the handbook.

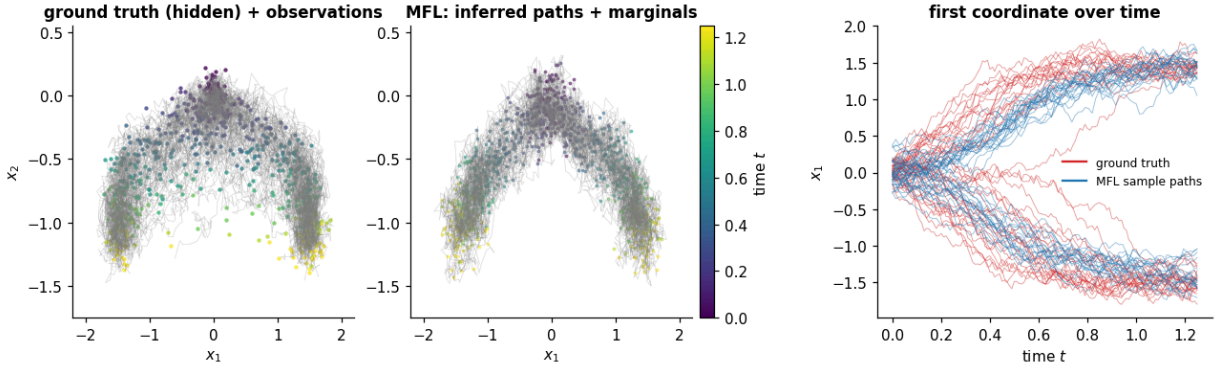


Figure 1: Trajectory inference on a simulated 10-dimensional bifurcating process (from the companion notebook). *Left:* hidden ground-truth paths (grey) and the observed snapshots, coloured by time. *Middle:* marginals and trajectories reconstructed by the min-entropy estimator, computed with mean-field Langevin dynamics. *Right:* first coordinate over time, ground truth vs. inferred sample paths.

2 Optimal transport essentials

This section fixes notation and recalls the facts used later; see (Peyré & Cuturi, 2019; Santambrogio, 2015; Villani, 2009) for details. Throughout, $c(x, y) = \frac{1}{2}\|x - y\|^2$ unless stated otherwise.

2.1 Static OT and duality

For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, the Kantorovich problem and its dual read

$$\mathrm{OT}(\mu, \nu) = \min_{\pi \in \Pi(\mu,\nu)} \int c \, \mathrm{d}\pi = \sup_{\varphi(x)+\psi(y)\le c(x,y)} \int \varphi \, \mathrm{d}\mu + \int \psi \, \mathrm{d}\nu. \tag{2}$$

At the optimum one may take $\psi = \varphi^c$, where $\varphi^c(y) = \inf_x c(x, y) - \varphi(x)$ is the c-transform. For the quadratic cost and μ absolutely continuous, Brenier's theorem gives a unique optimal plan $\pi = (\mathrm{id}, T)_{\#}\mu$ with $T = \nabla u$, $u = \frac{1}{2}\|\cdot\|^2 - \varphi$ convex. In particular $T(x) = x - \nabla\varphi(x)$, an identity that reappears in entropic form in Section 3.

2.2 Dynamic OT and the geometry of $\mathcal{P}_2$

Benamou and Brenier (Benamou & Brenier, 2000) showed that

$$W_2^2(\mu, \nu) = \min_{(\rho, v)} \int_0^1 \int \|v_t\|^2 \, \mathrm{d}\rho_t \, \mathrm{d}t \quad \text{s.t.} \quad \partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \quad \rho_0 = \mu, \, \rho_1 = \nu, \tag{3}$$

with $W_2^2 = 2\,\mathrm{OT}$. The minimiser is the *displacement interpolation* $\rho_t = ((1-t)\,\mathrm{id} + tT)_{\#}\mu$ (McCann, 1997): particles move on straight lines at constant speed and never cross.

Otto's formal Riemannian calculus (Ambrosio et al., 2008; Otto, 2001) reads Eq. 3 as a geodesic problem: tangent vectors at ρ are velocity fields $v = \nabla\varphi$, with metric $\|v\|_\rho^2 = \int \|v\|^2\,\mathrm{d}\rho$. The Wasserstein gradient of a functional $\mathcal{F}$ is $\nabla_{W_2}\mathcal{F}(\rho) = \nabla\frac{\delta\mathcal{F}}{\delta\rho}$, where $\frac{\delta\mathcal{F}}{\delta\rho}$ is the first variation, and the **Wasserstein gradient flow** is

$$\partial_t\rho_t = \nabla\cdot\left(\rho_t\nabla\frac{\delta\mathcal{F}}{\delta\rho}[\rho_t]\right). \tag{4}$$

Its time discretisation is the JKO scheme (Jordan et al., 1998), $\rho_{k+1} \in \arg\min_\rho W_2^2\frac{\rho,\rho_k}{2h} + \mathcal{F}(\rho)$.

Example 3 (Fokker–Planck as a gradient flow). For $\mathcal{F}(\rho) = \int\Psi\,\mathrm{d}\rho + \frac{\tau}{2}\int\rho\log\rho$, Eq. 4 is the Fokker–Planck equation $\partial_t\rho = \nabla\cdot(\rho\nabla\Psi) + \frac{\tau}{2}\Delta\rho$, i.e. the law of the Langevin diffusion $\mathrm{d}X_t = -\nabla\Psi(X_t)\,\mathrm{d}t + \sqrt{\tau}\,\mathrm{d}B_t$ (Jordan et al., 1998). If Ψ is λ-convex, $\mathcal{F}$ is λ-displacement convex (McCann, 1997) and the flow contracts in W_2 at rate λ, the starting point of the evolution-variational-inequality (EVI) theory of (Ambrosio et al., 2008).

Connection 4 (OT already is a model of trajectories). Displacement interpolation is a trajectory model: it reconstructs a deterministic, cyclically monotone flow between two snapshots. It is the **zero-noise** answer to TI. Its defects for biology are just as instructive: trajectories never cross, nothing is random, the interpolation between three or more snapshots is a broken geodesic with no memory of the dynamics, and the estimation of W_2 (and of its maps) from samples suffers from the curse of dimensionality, at rate $n^{-1/d}$. Each defect is cured by adding noise, which is where entropy enters.

3 Entropic OT and Sinkhorn's algorithm

3.1 Definition, dual and the Schrödinger system

Definition 5 (Entropic optimal transport). For $\varepsilon > 0$,

$$\mathrm{OT}_\varepsilon(\mu,\nu) = \min_{\pi\in\Pi(\mu,\nu)}\int c\,\mathrm{d}\pi + \varepsilon H(\pi\mid\mu\otimes\nu), \quad H(\pi\mid\rho) = \int\log\frac{\mathrm{d}\pi}{\mathrm{d}\rho}\,\mathrm{d}\pi. \tag{5}$$

Strict convexity gives a unique optimiser. Lagrangian duality yields an unconstrained concave dual (Nutz, 2021; Peyré & Cuturi, 2019):

$$\mathrm{OT}_\varepsilon(\mu,\nu) = \sup_{\varphi,\psi}\int\varphi\,\mathrm{d}\mu + \int\psi\,\mathrm{d}\nu - \varepsilon\int e^{(\varphi(x)+\psi(y)-c(x,y))/\varepsilon}\,\mathrm{d}\mu(x)\,\mathrm{d}\nu(y) + \varepsilon, \tag{6}$$

and the primal–dual relation $\mathrm{d}\pi^\star = e^{(\varphi\oplus\psi-c)/\varepsilon}\,\mathrm{d}(\mu\otimes\nu)$. The optimality conditions are the **Schrödinger system**

$$\varphi(x) = -\varepsilon\log\int e^{(\psi(y)-c(x,y))/\varepsilon}\,\mathrm{d}\nu(y), \quad \psi(y) = -\varepsilon\log\int e^{(\varphi(x)-c(x,y))/\varepsilon}\,\mathrm{d}\mu(x). \tag{7}$$

The right-hand sides are **soft c-transforms**: the softmin $-\varepsilon\log\int e^{-(\cdot)/\varepsilon}$ replaces the infimum in φ^c. As $\varepsilon\to 0$ they converge to the c-transform, and Eq. 6 to Eq. 2.

Sinkhorn's algorithm (Cuturi, 2013; Sinkhorn, 1964) alternates the two equations of Eq. 7. Equivalently, it is block-coordinate ascent on Eq. 6, or iterative proportional fitting: alternating KL projections of the Gibbs kernel $e^{-c/\varepsilon}\mu\otimes\nu$ onto the two marginal constraints. For m points each

iteration costs $O(m^2)$ and is embarrassingly parallel; in practice one runs it in the log domain for stability.

3.2 Three properties that matter for TI

Proposition 6 (Smooth potentials and barycentric projection). The Schrödinger potentials defined by Eq. 7 extend to smooth functions on all of $\mathcal{X}$, and for the quadratic cost

$$\nabla\varphi(x) = \int \nabla_x c(x,y)\,\mathrm{d}\pi^\star(y \mid x) = x - \mathbb{E}_{\pi^\star}[Y \mid X = x]. \tag{8}$$

Thus $x \mapsto x - \nabla\varphi(x)$ is the **barycentric projection** of the entropic plan, the entropic analogue of the Brenier map $T = \mathrm{id} - \nabla\varphi$. It is a statistically efficient estimator of T (Pooladian & Niles-Weed, 2021).

Proposition 7 (First variation). $\frac{\delta \mathrm{OT}_\varepsilon}{\delta\mu}(\mu,\nu) = \varphi$, the Schrödinger potential (up to an additive constant) (Feydy et al., 2019).

Combined with Eq. 8, this means that the **Wasserstein gradient of $\mu \mapsto \mathbf{OT}_\varepsilon(\mu,\nu)$ at a particle x points from x to the barycentre of where x is sent**. This is the force that couples the particle clouds in Section 6.

Proposition 8 (Statistics and limits). (i) As $\varepsilon \to 0$, $\mathrm{OT}_\varepsilon \to \mathrm{OT}$ and the optimal plans converge to an optimal transport plan (Carlier et al., 2017; Léonard, 2012). As $\varepsilon \to \infty$, $\pi^\star \to \mu \otimes \nu$. (ii) Plug-in estimation of OT_ε from n samples has error of order $n^{-1/2}$, independently of the dimension (with constants deteriorating as $\varepsilon \to 0$) (Genevay et al., 2019; Mena & Niles-Weed, 2019). Unregularised OT has the $n^{-1/d}$ rate.

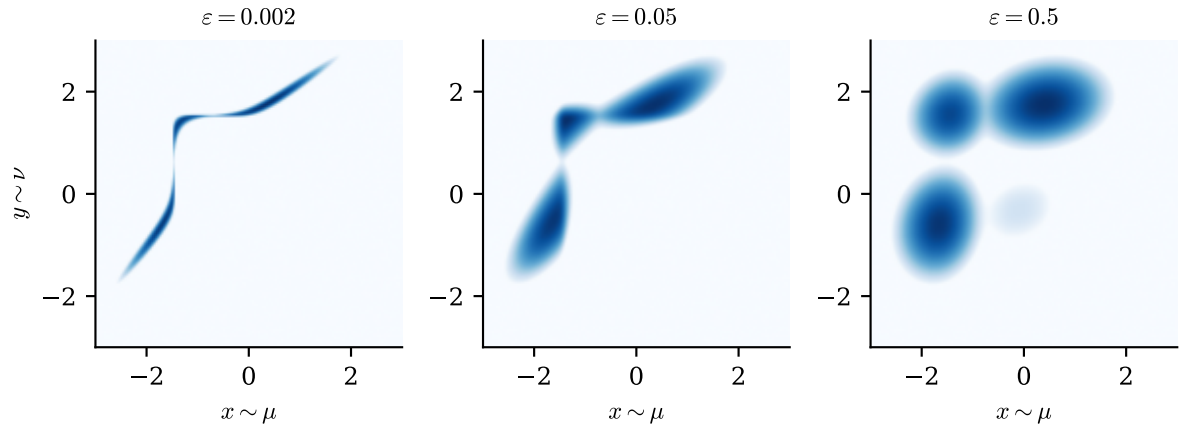


Figure 2: Entropic plans between two 1-D Gaussian mixtures (log-density). Small ε: nearly the monotone Brenier map. Large ε: approaching the independent coupling $\mu \otimes \nu$.

Connection 9 (Entropy is noise, not just a numerical trick). In machine learning, entropic regularisation is often introduced as a device to make OT smooth, fast and differentiable (Cuturi, 2013). For TI it has a *modelling* meaning: the entropic plan with $\varepsilon = \tau h$ is exactly the joint law at times 0 and h of Brownian particles with diffusivity τ, conditioned to start from μ and arrive at ν. This is Schrödinger's problem, the subject of the next section. The regularisation parameter is therefore not a tuning knob but the *noise level* of the biological process times the time gap. Moreover, the dimension-free sample complexity (ii) is the statistical reason why entropic methods remain usable with a few hundred cells in high dimension.

4 The Schrödinger problem

4.1 Schrödinger's thought experiment

In 1931–32 Schrödinger asked the following question (Léonard, 2014; Schrödinger, 1932). Release N independent Brownian particles with diffusivity ε. At time 0 their empirical distribution is close to μ, and at time 1 one observes, to one's surprise, an empirical distribution close to ν instead of the heat-flow prediction. **What is the most likely way the cloud got there?**

Let $R \in \mathcal{P}(\Omega)$ be the law of one reference Brownian path, and $L^N = \frac{1}{N}\sum_k \delta_{X^k}$ the empirical measure of the N paths. By Sanov's theorem, $\mathbb{P}(L^N \approx P) \approx \exp(-NH(P \mid R))$. Conditioning on the atypical event $\{L_0^N \approx \mu, L_1^N \approx \nu\}$, the empirical law of paths concentrates on the solution of

$$\min_{P \in \mathcal{P}(\Omega)} H(P \mid R) \quad \text{s.t.} \quad P_0 = \mu, P_1 = \nu. \tag{9}$$

This is the **dynamic Schrödinger problem**, and its solution the **Schrödinger bridge** (Föllmer, 1988; Léonard, 2014).

Connection 10 (The estimator as a conditional large-deviation principle). Read with biological eyes, Eq. 9 is a statement about trajectory inference. Under the null model "cells are independent Brownian particles", the most likely population-level explanation of two observed snapshots is the Schrödinger bridge. The min-entropy estimator of Section 5 extends exactly this principle to many noisy snapshots.

4.2 From dynamic to static: the bridge decomposition

Relative entropy decomposes along the endpoint map: for any P,

$$H(P \mid R) = H(P_{01} \mid R_{01}) + \int H(P^{xy} \mid R^{xy}) \, \mathrm{d}P_{01}(x, y), \tag{10}$$

where P^{xy} and R^{xy} are the laws conditioned on $X_0 = x, X_1 = y$. The second term vanishes iff P uses the reference bridges, $P = \int R^{xy} \, \mathrm{d}P_{01}$. Hence Eq. 9 reduces to the **static Schrödinger problem** $\min_{\pi \in \Pi(\mu,\nu)} H(\pi \mid R_{01})$, and the dynamic solution is obtained by gluing Brownian bridges onto the static plan.

Proposition 11 (Static Schrödinger = entropic OT). Let R be Brownian motion with diffusivity ε started from Lebesgue measure, so that $\mathrm{d}R_{01}(x, y) \propto e^{-\|x-y\|^2/(2\varepsilon)} \, \mathrm{d}x \, \mathrm{d}y$. For $\pi \in \Pi(\mu, \nu)$,

$$\varepsilon H(\pi \mid R_{01}) = \int \frac{1}{2}\|x - y\|^2 \, \mathrm{d}\pi + \varepsilon H(\pi \mid \mu \otimes \nu) + \varepsilon(H(\mu) + H(\nu)) + \text{const}, \tag{11}$$

with $H(\mu) = \int \log \mu \, \mathrm{d}\mu$. Hence the static Schrödinger problem and $\mathrm{OT}_\varepsilon(\mu, \nu)$ have the same minimiser.

Proof sketch. Write $\frac{\mathrm{d}\pi}{\mathrm{d}R_{01}} = \frac{\mathrm{d}\pi}{\mathrm{d}(\mu\otimes\nu)} \cdot \frac{\mathrm{d}(\mu\otimes\nu)}{\mathrm{d}x\,\mathrm{d}y} \cdot \frac{\mathrm{d}x\,\mathrm{d}y}{\mathrm{d}R_{01}}$ and integrate the logarithm against π; the middle factor contributes $H(\mu) + H(\nu)$, the last one $\|x - y\|^2/(2\varepsilon)$ plus a constant. □

On a manifold, or with a non-Gaussian reference, the cost becomes $c_\varepsilon = -\varepsilon \log p_\varepsilon$, with p_ε the heat kernel. By Varadhan's formula $c_\varepsilon \to d^2/2$ as $\varepsilon \to 0$, which is the cost used in the TI papers.

4.3 Stochastic control and the fluid picture

Let R^ε be the law of $\mathrm{d}X_t = \sqrt{\varepsilon}\,\mathrm{d}B_t$. If $H(P \mid R^\varepsilon) < \infty$, then under P the canonical process solves $\mathrm{d}X_t = b_t(X_t)\,\mathrm{d}t + \sqrt{\varepsilon}\,\mathrm{d}B_t$ for some drift b, and by Girsanov's theorem (Föllmer, 1988)

$$H(P \mid R^\varepsilon) = H(P_0 \mid R_0^\varepsilon) + \frac{1}{2\varepsilon}\mathbb{E}_P \int_0^1 \|b_t(X_t)\|^2 \, \mathrm{d}t. \tag{12}$$

The Schrödinger problem is therefore a **stochastic optimal control** problem, which is to steer a diffusion from μ to ν with minimal kinetic energy of the control (Chen et al., 2016; 2021; Dai Pra, 1991). Replacing the drift by the *current velocity* $v_t = b_t - \frac{\varepsilon}{2}\nabla \log \rho_t$, which satisfies the continuity equation $\partial_t \rho + \nabla \cdot (\rho v) = 0$, and integrating the cross term (it is a time derivative of $\int \rho_t \log \rho_t$) gives the following fluid formulation (Chen et al., 2016; Gentil et al., 2017).

Theorem 12 (Entropic Benamou–Brenier formula). Up to the additive constant $\frac{\varepsilon}{2}(H(\nu) - H(\mu))$, the value of Eq. 9 times ε equals

$$\min_{(\rho,v)} \int_0^1 \int \left(\frac{1}{2}\|v_t\|^2 + \frac{\varepsilon^2}{8}\|\nabla \log \rho_t\|^2 \right) \mathrm{d}\rho_t \, \mathrm{d}t \quad \text{s.t.} \quad \partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0,\ \rho_0 = \mu,\ \rho_1 = \nu. \tag{13}$$

The second term is $\varepsilon^2/8$ times the **Fisher information**, so the Schrödinger bridge is dynamic OT with a penalty on the concentration of the interpolating densities. As $\varepsilon \to 0$, Eq. 13 Γ-converges to Eq. 3, the Schrödinger bridges converge to the displacement interpolation, and the static plans to the Brenier plan (Léonard, 2012; Mikami, 2004). Much of displacement convexity theory has an "entropic interpolation" analogue (Gentil et al., 2017).

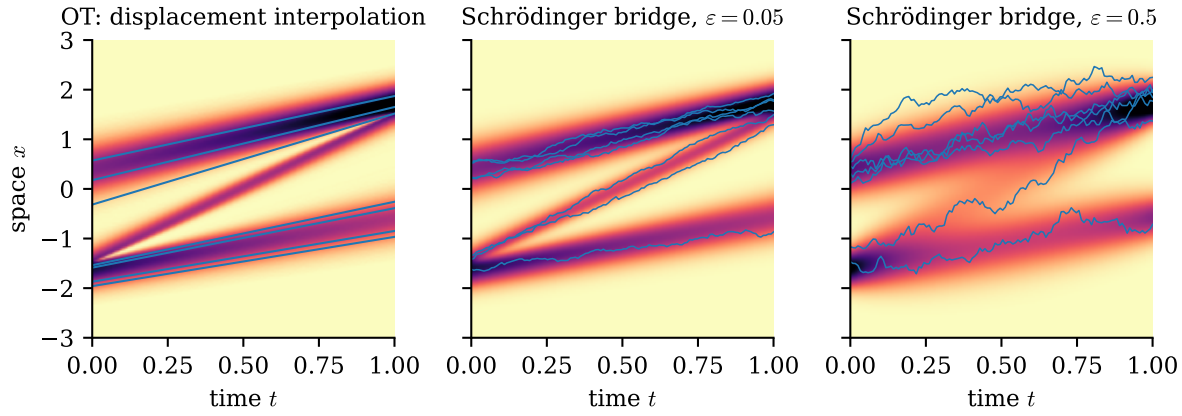


Figure 3: Displacement interpolation vs. Schrödinger bridges between the same 1-D marginals (space-time densities, with a few sample paths). The bridge is a mixture of Brownian bridges glued to the entropic plan, $\rho_t = \int \mathcal{N}((1-t)x + ty, \varepsilon t(1-t))\, \mathrm{d}\pi^\star(x,y)$, so it widens in the middle of the interval and its paths cross. As $\varepsilon \to 0$ it collapses onto the geodesic.

4.4 Dictionary

	Optimal transport ($\varepsilon = 0$)	**Schrödinger / entropic OT** ($\varepsilon > 0$)
Optimal plan	deterministic map $(\mathrm{id}, \nabla u)_{\#}\mu$	Gibbs plan $e^{(\varphi \oplus \psi - c)/\varepsilon}\mu \otimes \nu$
Dual operator	c-transform (inf)	soft c-transform (softmin)
Optimality	Monge–Ampère equation	Schrödinger system Eq. 7
Map	Brenier map $x - \nabla\varphi(x)$	barycentric projection $\mathbb{E}[Y \mid X = x]$
Interpolation	straight lines, McCann	Brownian bridges, entropic interpolation
Dynamic form	Benamou–Brenier Eq. 3	Eq. 13 (+ Fisher information) / stochastic control
Algorithm	LP, auction, semi-discrete Newton	Sinkhorn, $O(m^2)$ per iteration, GPU-friendly
Sample complexity	$n^{-1/d}$	$n^{-1/2}$ (constants exponential in $1/\varepsilon$)
Probabilistic meaning	—	most likely evolution of Brownian particles (Sanov)

Table 2: Correspondences between OT and the Schrödinger problem.

5 Trajectory inference as a multi-marginal Schrödinger problem

5.1 The population model and why it is a gradient flow

The standard model for a population without births or deaths is the drift-diffusion

$$\mathrm{d}X_t = -\nabla\Psi(t, X_t)\,\mathrm{d}t + \sqrt{\tau}\,\mathrm{d}B_t, \quad X_0 \sim \mu_0, \tag{14}$$

where Ψ is the (unknown) **Waddington landscape** (Hashimoto et al., 2016; Waddington, 1957) and $\tau > 0$ a known diffusivity. Only the gradient part of a drift is identifiable from marginals: a divergence-free component that leaves ρ_t invariant changes the trajectories but not the snapshots (Lavenant et al., 2021; Weinreb et al., 2018). This is why one assumes a gradient drift.

Connection 13 (Cells follow a Wasserstein gradient flow). By the Fokker–Planck example of Section 2, the marginals of Eq. 14 are the Wasserstein gradient flow of the free energy $\mathcal{F}_t(\rho) = \int \Psi(t,\cdot)\,\mathrm{d}\rho + \frac{\tau}{2}\int \rho\log\rho$ (non-autonomous if Ψ depends on t). Recovering the dynamics is therefore an **inverse problem for a gradient flow**: given a few points along a gradient-flow curve, recover the energy (the landscape) and the stochastic trajectories. Parametric approaches such as JKOnet (Bunne et al., 2022) take this literally and fit Ψ_θ so that JKO steps map each snapshot to the next.

5.2 The min-entropy estimator

Let W^τ be the law of (reflected) Brownian motion with diffusivity τ. Lavenant et al. (2021) define

$$R^\star = \operatorname*{arg\,min}_{R \in \mathcal{P}(\Omega)} \frac{1}{\lambda}\sum_{i=1}^{T} \Delta t_i\, \mathrm{Fit}_\sigma\big(R_{t_i}, \hat{\mu}_i\big) + \tau H(R \mid W^\tau), \tag{15}$$

with quadrature weights Δt_i, a regularisation $\lambda > 0$ and a data-fitting functional, e.g. the negative log-likelihood under Gaussian observation noise used in (Chizat et al., 2022):

$$\mathrm{Fit}_\sigma(\rho, \hat{\mu}) = -\int \log\left(\int e^{-\|x-y\|^2/(2\sigma^2)}\,\mathrm{d}\rho(x)\right)\mathrm{d}\hat{\mu}(y). \tag{16}$$

This is a **soft-constrained, multi-marginal Schrödinger problem**. With hard constraints $R_{t_i} = \mu_i$ it would be the Schrödinger problem with T prescribed marginals; the fitting term relaxes these constraints because the $\hat{\mu}_i$ are noisy, finite samples.

Theorem 14 (Consistency; Lavenant et al. (2021)). If the ground truth is of the form Eq. 14 (on a compact manifold, with suitable regularity), and the measurement times become dense in $[0,1]$, then $R^\star \to P$ as $T \to \infty$ followed by $\lambda, \sigma \to 0$, weakly and almost surely.

Why does minimising entropy recover a *drift*? By Eq. 12, $\tau H(R \mid W^\tau)$ is, up to the initial term, the kinetic energy $\frac{1}{2}\mathbb{E}\int \|b_t\|^2\,\mathrm{d}t$ of the drift. The estimator is thus a **least-action principle**: among all drift-diffusions with noise τ that explain the snapshots, choose the one with the least kinetic energy. For gradient drifts that are densely observed, the least-action explanation is the truth.

5.3 The representer theorem: from paths to T marginals

Because W^τ is Markov, Eq. 10 generalises to a chain. For any R with marginals $\mu_i = R_{t_i}$,

$$H(R \mid W^\tau) \geq \sum_{i=1}^{T-1} H\left(R_{t_i,t_{i+1}} \mid p_{\tau_i}\mu_i \otimes \mu_{i+1}\right) + \sum_{i=1}^{T} H(\mu_i) + C, \quad \tau_i = \tau(t_{i+1} - t_i), \tag{17}$$

with equality iff R is Markov and uses Brownian bridges between the observation times (Chizat et al., 2022). Optimising over the couplings with marginals fixed yields entropic OT terms, and we arrive at the **reduced problem** over $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_T) \in \mathcal{P}(\mathcal{X})^T$:

$$F(\boldsymbol{\mu}) = \underbrace{\frac{1}{\lambda}\sum_i \Delta t_i\,\mathrm{Fit}_\sigma(\mu_i, \hat{\mu}_i) + \sum_{i=1}^{T-1}\frac{\mathrm{OT}_{\tau_i}(\mu_i, \mu_{i+1})}{t_{i+1} - t_i}}_{G(\boldsymbol{\mu})} + \tau\sum_{i=1}^{T} H(\mu_i). \tag{18}$$

Theorem 15 (Representer theorem; Chizat et al. (2022)). F is jointly convex with a unique minimiser $\boldsymbol{\mu}^\star$, and $R^\star = \int W^\tau(\cdot \mid x_1, \ldots, x_T)\,\mathrm{d}\left(\gamma_{1,2}\gamma_{2,3}\cdots\gamma_{T-1,T}\right)(x_1, \ldots, x_T)$, where $\gamma_{i,i+1}$ are the optimal entropic plans between $\mu_i^\star$ and $\mu_{i+1}^\star$. In words: optimal marginals, chained by Schrödinger couplings, filled in with Brownian bridges.

Connection 16 (Three readings of the reduced objective).

1. **Waddington-OT as a limit.** If the marginals are pinned to the data ($\lambda \to 0$ with exact marginals), F reduces to a sum of pairwise entropic OT problems between consecutive snapshots: exactly Waddington-OT (Schiebinger et al., 2019). WOT is the *interpolating* special case; the min-entropy estimator is its *smoothing* (regression) version, which also corrects the marginals.
2. **Multi-marginal OT with a path-structured cost.** Equivalently, $R_{t_1,\ldots,t_T}$ solves an entropic multi-marginal problem (regularisation τ) with cost $\sum_i \|x_{i+1} - x_i\|^2/(2(t_{i+1} - t_i))$. Tree-structured costs keep multi-marginal Sinkhorn tractable (Haasler et al., 2021).
3. **A smoothing spline in Wasserstein space.** $\sum_i \mathrm{OT}_{\tau_i}(\mu_i, \mu_{i+1})/\Delta t$ is a discrete, entropic version of the action $\int \|v\|^2\,\mathrm{d}\rho\,\mathrm{d}t$ in Eq. 3. So Eq. 18 is a data term plus a *first-order* (piecewise geodesic) roughness penalty: a smoothing spline on $(\mathcal{P}_2, W_2)$, blurred by noise. Second-order (cubic) splines on Wasserstein space (Benamou et al., 2019; Chewi et al., 2021) would penalise acceleration instead. They form a natural, largely unexplored alternative prior for TI.

Note the **positive** sign of $\tau \sum H(\mu_i)$ in Eq. 18: the marginal entropies penalise concentration, which is what makes F jointly convex even though the coupling terms are only separately convex (Chizat et al., 2022).

6 Two gradient flows: mean-field Langevin dynamics

6.1 Two clocks

There are two unrelated "times" in this story, and two gradient flows:

	Physical time t	**Optimisation time s**
Space	$\mathcal{P}(\mathcal{X})$	$\mathcal{P}(\mathcal{X})^T$ (one cloud per snapshot)
Energy	free energy $\int \Psi \, \mathrm{d}\rho + \frac{\tau}{2} H(\rho)$ (unknown)	reduced objective F of Eq. 18 (known)
Flow	Fokker–Planck of the cells	mean-field Langevin dynamics
Role	the model to be inferred	the algorithm that infers it

Table 3: The two gradient flows in trajectory inference.

6.2 Mean-field Langevin dynamics

For a functional of the form $F(\mu) = G(\mu) + \tau H(\mu)$, the Wasserstein gradient flow Eq. 4 is

$$\partial_s \mu_s = \nabla \cdot \left(\mu_s \nabla \frac{\delta G}{\delta \mu}[\mu_s] \right) + \tau \Delta \mu_s, \tag{19}$$

the law of the McKean–Vlasov diffusion $\mathrm{d}X_s = -\nabla \frac{\delta G}{\delta \mu}[\mu_s](X_s)\,\mathrm{d}s + \sqrt{2\tau}\,\mathrm{d}B_s$, $\mu_s = \mathrm{Law}(X_s)$. This is **mean-field Langevin (MFL) dynamics**. It was introduced to study wide two-layer neural networks (Hu et al., 2021; Mei et al., 2018), and it is simulated with m interacting particles and noisy gradient steps.

For TI, Chizat et al. (2022) apply MFL on the product space $\mathcal{P}(\mathcal{X})^T$, one species per snapshot. Using the first variations of Section 3, the potential acting on species i is

$$V^{(i)}[\boldsymbol{\mu}] = \frac{\delta \, \mathrm{Fit}}{\delta \mu_i} + \frac{\varphi_{i,i+1}}{t_{i+1} - t_i} + \frac{\psi_{i-1,i}}{t_i - t_{i-1}}, \tag{20}$$

where $(\varphi_{i,i+1}, \psi_{i,i+1})$ are the Schrödinger potentials of $\mathrm{OT}_{\tau_i}(\mu_i, \mu_{i+1})$. By Eq. 8 each particle feels three forces: a soft-min attraction to nearby observations at t_i, a pull towards the barycentre of its expected successors at t_{i+1}, and a pull towards the barycentre of its expected predecessors at t_{i-1}. The T clouds behave like **beads on an elastic chain in Wasserstein space**, kept apart by diffusion and pinned by the data. One step of the particle scheme costs one warm-started Sinkhorn solve per pair of consecutive clouds.

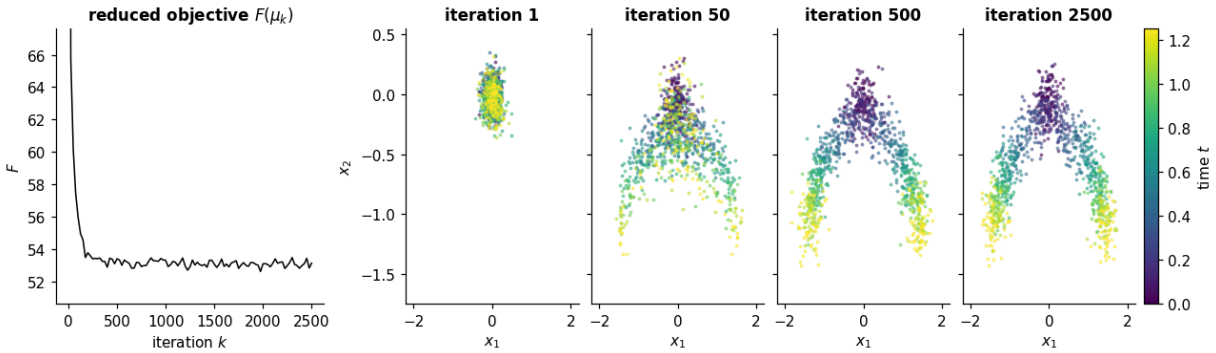


Figure 4: MFL on the bifurcation example (companion notebook). *Left:* reduced objective F along the iterations (entropy estimated by k-NN). *Right:* particle clouds at iterations 1, 50, 500, 2500, coloured by physical time t.

6.3 Why it converges: flat convexity meets Wasserstein dissipation

The convergence theory of MFL is instructive for a gradient-flow audience, because it does **not** rely on displacement convexity. Let $\hat{\mu} \propto e^{-\frac{\delta G}{\delta \mu}[\mu]/\tau}$ be the **proximal Gibbs measure** of μ. Its fixed points are exactly the minimisers of F.

Lemma 17 (Entropy sandwich). If G is convex along linear (flat) interpolations $(1-s)\mu + s\nu$, then $F(\mu) - \min F \le \tau H(\mu \mid \hat{\mu})$.

Proof sketch. By flat convexity, $G(\nu) \ge G(\mu) + \int \frac{\delta G}{\delta \mu}[\mu]\, \mathrm{d}(\nu - \mu)$. Adding $\tau H(\nu)$ and minimising over ν with the Gibbs variational principle $\min_\nu \int V \,\mathrm{d}\nu + \tau H(\nu) = -\tau \log \int e^{-V/\tau}$ gives a lower bound on $\min F$. Comparing with $F(\mu)$ yields exactly $\tau H(\mu \mid \hat{\mu})$. □

Along Eq. 19, the energy dissipates at the rate of the relative Fisher information,

$$\frac{\mathrm{d}}{\mathrm{d}s} F(\mu_s) = -\int \left\| \nabla \frac{\delta F}{\delta \mu} \right\|^2 \mathrm{d}\mu_s = -\tau^2 I(\mu_s \mid \hat{\mu}_s). \tag{21}$$

If the proximal Gibbs measures satisfy a **log-Sobolev inequality** with a uniform constant C_{LS}, i.e. $H(\mu \mid \hat{\mu}) \le \frac{1}{2C_{\mathrm{LS}}} I(\mu \mid \hat{\mu})$, then combining the two displays gives

$$\frac{\mathrm{d}}{\mathrm{d}s}(F(\mu_s) - \min F) \le -2C_{\mathrm{LS}}\tau (F(\mu_s) - \min F), \tag{22}$$

hence $F(\mu_s) - \min F \le e^{-2C_{\mathrm{LS}}\tau s}(F(\mu_0) - \min F)$. This is the argument of (Chizat, 2022; Nitanda et al., 2022).

Theorem 18 (Global convergence for TI; Chizat et al. (2022)). On the torus, with an extra noise $\varepsilon > 0$ (MFL at temperature $\tau + \varepsilon$), the dynamics converges exponentially to the unique minimiser of $F_\varepsilon = G + (\tau + \varepsilon)H$, at a rate $C = \beta e^{-\alpha/\varepsilon}$. With a slowly annealed $\varepsilon_s \to 0$ it converges to $\boldsymbol{\mu}^\star$, and thus, by the representer theorem, to the min-entropy estimator.

Why the extra noise ε? In TI the smooth part G is only *separately* convex in each μ_i; it is $G + \tau H$ that is jointly convex (the positive marginal entropies make it so). The argument above is therefore applied to the convex functional $G + \tau H$ with εH playing the role of the entropy: the temperature in the LSI is ε, not τ. The uniform LSI comes from Holley–Stroock perturbation (Holley & Stroock, 1987). The oscillation of $V^{(i)}$ is bounded uniformly, by the oscillation bound for Schrödinger potentials and the boundedness of the fitting gradient, so the constant degrades like $e^{-\mathrm{osc}(V)/\varepsilon}$. This is the price of non-convex landscapes, familiar from simulated annealing.

Connection 19 (Convexity in one geometry, dissipation in another). Classical gradient-flow theory (Ambrosio et al., 2008) pairs a metric with convexity *in the same metric*: λ-displacement convexity yields EVI, contraction and rates. Here the objective is convex along *linear* interpolations (for TI: $G + \tau H$, built from soft-min fits and entropic OT terms), and nothing is known about its displacement convexity. MFL pairs flat convexity, which controls the optimality gap (sandwich lemma), with Wasserstein dissipation, which controls its decrease (Fisher information). The two are linked by a functional inequality (LSI), much as in the Otto–Villani theorem linking LSI and transport inequalities (Otto & Villani, 2000). This mixed-geometry mechanism is the conceptual core of the guarantee, and it is a natural place to bring other geometries into play (Section 7).

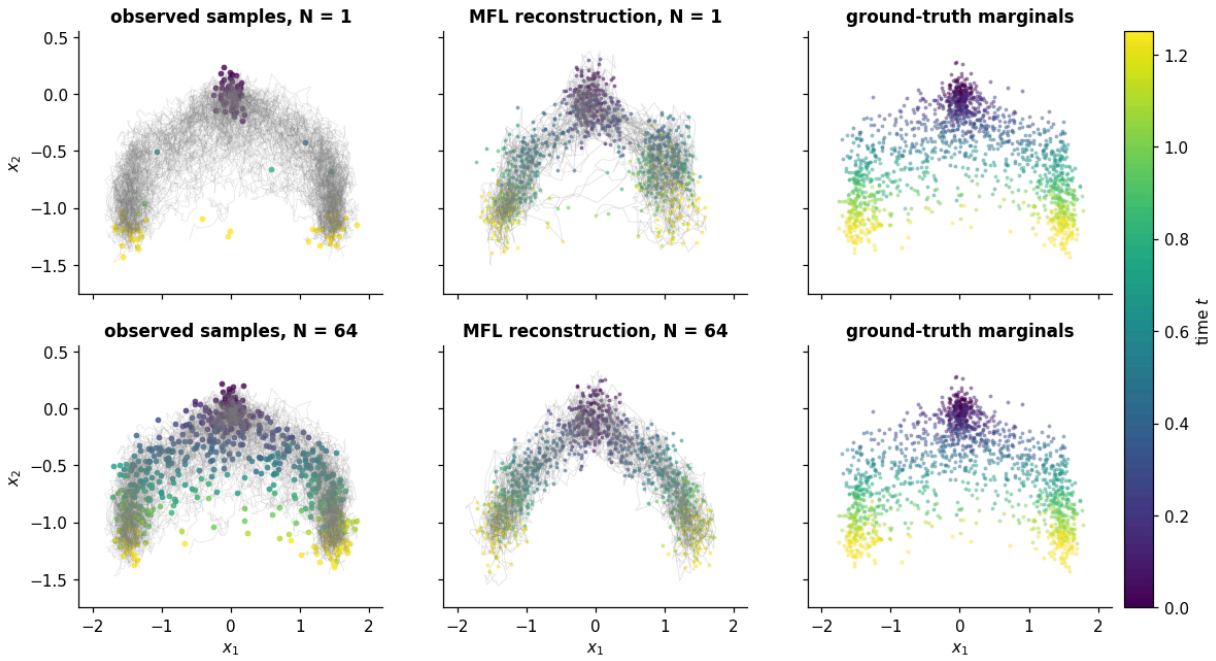


Figure 5: Grid-free reconstruction from scarce data (companion notebook). *Top:* one observed cell per intermediate time point. *Bottom:* 64 cells. MFL particles move freely in $\mathcal{X}$, so information propagates from well-sampled time points through the Schrödinger couplings. A fixed-support discretisation can only place mass on observed points.

7 Mass is not conserved: unbalanced transport and growth

7.1 Growth in the population model

Cells divide and die. With a (state- and time-dependent) net growth rate g, the population density satisfies a reaction–drift–diffusion equation

$$\partial_t \rho = \nabla \cdot (\rho \nabla \Psi) + \frac{\tau}{2} \Delta \rho + g\rho. \tag{23}$$

Snapshots are normalised, so they show the **relative** abundance of states. A fast-dividing subpopulation looks like a population that cells migrate *into*. Conflating proliferation with migration is the central failure mode that unbalanced methods address.

7.2 Unbalanced optimal transport

Unbalanced OT relaxes mass conservation. In dynamic form, the **Wasserstein–Fisher–Rao** (WFR), or **Hellinger–Kantorovich**, distance (Chizat et al., 2018a; Kondratyev et al., 2016; Liero et al., 2018) allows a source term,

$$\mathrm{WFR}^2(\mu,\nu) = \min_{(\rho,v,\alpha)} \int_0^1 \int \left(\|v_t\|^2 + \kappa\, \alpha_t^2 \right) \mathrm{d}\rho_t \, \mathrm{d}t \quad \text{s.t.} \quad \partial_t \rho + \nabla \cdot (\rho v) = \rho \alpha, \tag{24}$$

with a length scale $\kappa > 0$ (normalisations differ between references). Its static counterpart is the optimal entropy-transport problem

$$\min_{\gamma \in \mathcal{M}_+(\mathcal{X}^2)} \int c \, \mathrm{d}\gamma + \rho_{\mathrm{m}} H(\gamma_1 \mid \mu) + \rho_{\mathrm{m}} H(\gamma_2 \mid \nu), \tag{25}$$

in which the hard marginal constraints are replaced by KL penalties of strength ρ_{m} (Liero et al., 2018). With an additional entropic term this is solved by **scaling algorithms**, i.e. Sinkhorn iterations

with damped updates (Chizat et al., 2018b). Waddington-OT uses precisely this, with the marginals reweighted by a prior on cellular growth rates (Schiebinger et al., 2019).

The TI paper (Chizat et al., 2022) incorporates growth heuristically. Given a prior g, the Schrödinger bridge between consecutive clouds uses reweighted marginals, the source μ_i multiplied by $e^{g\Delta t/2}$ and the target μ_{i+1} by $e^{-g\Delta t/2}$, plus KL-relaxed marginal constraints. (The sign convention given here is the one consistent with growth and used in the authors' code.)

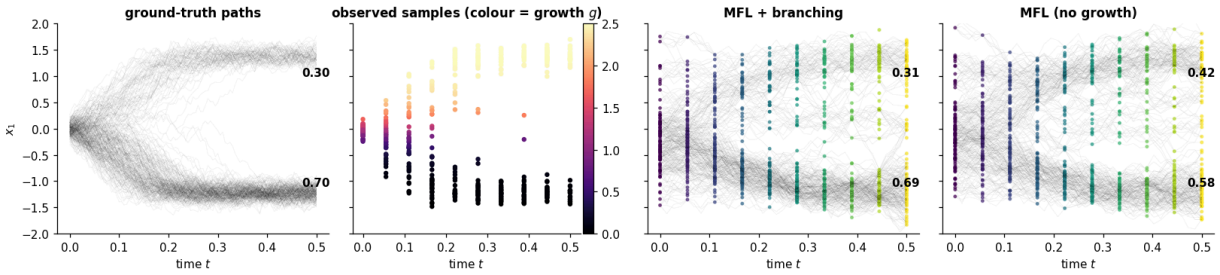


Figure 6: Growth confounds snapshots (companion notebook). The upper branch divides faster, so later snapshots are dominated by it, although 70% of lineages go down. MFL with the growth-reweighted bridges recovers 0.69. Without growth, mass conservation forces a constant branch fraction and the paths over-populate the upper branch.

7.3 The gradient-flow view of growth

In Otto's picture, the Fisher–Rao (Hellinger) geometry has gradient flows of **birth–death** type,

$$\partial_s \mu = -\mu\left(\frac{\delta F}{\delta \mu} - \int \frac{\delta F}{\delta \mu}\,\mathrm{d}\mu\right), \tag{26}$$

which teleport mass instead of transporting it. The WFR gradient flow sums the transport (Wasserstein) and reaction (Fisher–Rao) parts. For MFL-type objectives, adding birth–death moves to Langevin dynamics is known to accelerate convergence, in particular across energy barriers where pure diffusion is metastable (Lu et al., 2019; Rotskoff et al., 2019).

Connection 20 (Growth suggests a WFR mean-field Langevin dynamics). Two independent reasons point to the same geometry. *Modelling*: with growth, the marginals of Eq. 23 live in $\mathcal{M}_+(\mathcal{X})$, and the natural reference process is branching Brownian motion, whose entropic transport problems are unbalanced. *Optimisation*: the MFL particles of different branches must exchange mass. Under pure Wasserstein dynamics this requires crossing low-density regions, which is exponentially slow in $1/\varepsilon$ (the $e^{-\alpha/\varepsilon}$ in the rate above), whereas Fisher–Rao moves would reweight particles directly. A principled formulation of the min-entropy estimator for branching processes, together with a WFR / Hellinger–Kantorovich gradient flow to compute it and a corresponding "mixed-geometry" convergence analysis, is a natural open problem at the intersection of TI and the theory of unbalanced gradient flows.

8 The wider landscape

The min-entropy estimator is *nonparametric* and *convex*. Much of the recent literature trades these properties for scalability or structure.

- **Parametric landscapes.** Fit Ψ_θ (or a drift network) so that simulated marginals match the snapshots: generative RNNs (Hashimoto et al., 2016), PRESCIENT (Yeo et al., 2021), JKOnet, which learns the energy of a JKO scheme (Bunne et al., 2022), and TrajectoryNet, a continuous normalising flow regularised by the Benamou–Brenier action (Tong et al., 2020). These approaches gain a structural prior and the ability to extrapolate, at the cost of non-convex

training and no global guarantees. The paper's conclusion suggests learning Ψ *jointly* with R as a hybrid.

- **Neural Schrödinger bridges.** Iterative proportional fitting in path space (diffusion Schrödinger bridges (De Bortoli et al., 2021)) and bridge matching (Shi et al., 2023) scale the dynamic problem Eq. 9 to high dimension with neural drifts. Flow matching (Lipman et al., 2023) with mini-batch OT couplings (Tong et al., 2024) is its $\varepsilon \to 0$, simulation-free relative.
- **Closed forms.** Between Gaussians, both entropic OT and the Schrödinger bridge are explicit (Bunne et al., 2023; Janati et al., 2020). They are useful as reference processes and as sanity checks.

9 Open problems for the OT and gradient-flow community

1. **Statistical rates.** Consistency is known (Lavenant et al., 2021); quantitative rates in terms of T, n_i, d, λ, σ, τ are largely open. The dimension-free sample complexity of entropic OT suggests favourable rates in the number of cells.
2. **Geometry of F.** Is the reduced objective (geodesically) convex along entropic or displacement interpolations in $\mathcal{P}(\mathcal{X})^T$? A positive answer would bring EVI-type contraction to TI.
3. **Discrete-time and finite-particle guarantees.** Uniform-in-time propagation of chaos and step-size analysis for the particle scheme with inexact Sinkhorn steps.
4. **Unbalanced min-entropy estimation.** Entropy relative to branching Brownian motion, its reduction to unbalanced Schrödinger bridges, and a WFR mean-field Langevin solver with global guarantees (Section 7).
5. **Better priors.** Learning Ψ jointly with the path law, non-Brownian reference processes (e.g. Ornstein–Uhlenbeck or learned references), and second-order (spline) priors in Wasserstein space (Benamou et al., 2019; Chewi et al., 2021).
6. **Beyond Euclidean state spaces.** Manifold-valued cell states and costs given by learned metrics, where Varadhan's formula links the reference heat kernel to the transport cost.

A companion Jupyter notebook implements everything in this handbook from scratch (Sinkhorn, MFL, path sampling, branching) and reproduces the figures; see the repository root.

References

Ambrosio, L., Gigli, N., & Savaré, G. (2008). *Gradient Flows in Metric Spaces and in the Space of Probability Measures* (2nd ed.). Birkhäuser.

Benamou, J.-D., & Brenier, Y. (2000). A Computational Fluid Mechanics Solution to the Monge–Kantorovich Mass Transfer Problem. *Numerische Mathematik*, *84*(3), 375–393.

Benamou, J.-D., Gallouët, T. O., & Vialard, F.-X. (2019). Second-Order Models for Optimal Transport and Cubic Splines on the Wasserstein Space. *Foundations of Computational Mathematics*, *19*(5), 1113–1143.

Bunne, C., Hsieh, Y.-P., Cuturi, M., & Krause, A. (2023). The Schrödinger Bridge Between Gaussian Measures Has a Closed Form. *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*.

Bunne, C., Papaxanthos, L., Krause, A., & Cuturi, M. (2022). Proximal Optimal Transport Modeling of Population Dynamics. *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, 6511–6528.

Carlier, G., Duval, V., Peyré, G., & Schmitzer, B. (2017). Convergence of Entropic Schemes for Optimal Transport and Gradient Flows. *SIAM Journal on Mathematical Analysis*, *49*(2), 1385–1418.

Chen, Y., Georgiou, T. T., & Pavon, M. (2016). On the Relation Between Optimal Transport and Schrödinger Bridges: A Stochastic Control Viewpoint. *Journal of Optimization Theory and Applications*, *169*(2), 671–691.

Chen, Y., Georgiou, T. T., & Pavon, M. (2021). Stochastic Control Liaisons: Richard Sinkhorn Meets Gaspard Monge on a Schrödinger Bridge. *SIAM Review*, *63*(2), 249–313.

Chewi, S., Clancy, J., Le Gouic, T., Rigollet, P., Stepaniants, G., & Stromme, A. (2021). Fast and Smooth Interpolation on Wasserstein Space. *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics*.

Chizat, L. (2022). Mean-Field Langevin Dynamics: Exponential Convergence and Annealing. *Transactions on Machine Learning Research*.

Chizat, L., Peyré, G., Schmitzer, B., & Vialard, F.-X. (2018a). An Interpolating Distance Between Optimal Transport and Fisher–Rao Metrics. *Foundations of Computational Mathematics*, *18*(1), 1–44.

Chizat, L., Peyré, G., Schmitzer, B., & Vialard, F.-X. (2018b). Scaling Algorithms for Unbalanced Optimal Transport Problems. *Mathematics of Computation*, *87*(314), 2563–2609.

Chizat, L., Zhang, S., Heitz, M., & Schiebinger, G. (2022). Trajectory Inference via Mean-field Langevin in Path Space. *Advances in Neural Information Processing Systems*, *35*.

Cuturi, M. (2013). Sinkhorn Distances: Lightspeed Computation of Optimal Transport. *Advances in Neural Information Processing Systems*, *26*.

Dai Pra, P. (1991). A Stochastic Control Approach to Reciprocal Diffusion Processes. *Applied Mathematics and Optimization*, *23*(1).

De Bortoli, V., Thornton, J., Heng, J., & Doucet, A. (2021). Diffusion Schrödinger Bridge with Applications to Score-Based Generative Modeling. *Advances in Neural Information Processing Systems*, *34*.

Feydy, J., Séjourné, T., Vialard, F.-X., Amari, S.-i., Trouvé, A., & Peyré, G. (2019). Interpolating Between Optimal Transport and MMD Using Sinkhorn Divergences. *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics*, 2681–2690.

Föllmer, H. (1988). Random Fields and Diffusion Processes. *École D'été De Probabilités De Saint-Flour XV–XVII, 1985–87, Lecture Notes in Mathematics 1362*, 101–203.

Genevay, A., Chizat, L., Bach, F., Cuturi, M., & Peyré, G. (2019). Sample Complexity of Sinkhorn Divergences. *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics*, 1574–1583.

Gentil, I., Léonard, C., & Ripani, L. (2017). About the Analogy Between Optimal Transport and Minimal Entropy. *Annales De La Faculté Des Sciences De Toulouse: Mathématiques*, *26*(3), 569–600.

Haasler, I., Ringh, A., Chen, Y., & Karlsson, J. (2021). Multimarginal Optimal Transport with a Tree-Structured Cost and the Schrödinger Bridge Problem. *SIAM Journal on Control and Optimization*, *59*(4).

Hashimoto, T., Gifford, D., & Jaakkola, T. (2016). Learning Population-Level Diffusions with Generative RNNs. *Proceedings of the 33rd International Conference on Machine Learning*, 2417–2426.

Holley, R., & Stroock, D. (1987). Logarithmic Sobolev Inequalities and Stochastic Ising Models. *Journal of Statistical Physics*, *46*(5), 1159–1194.

Hu, K., Ren, Z., Šiška, D., & Szpruch, Ł. (2021). Mean-Field Langevin Dynamics and Energy Landscape of Neural Networks. *Annales De L'institut Henri Poincaré (B) Probabilités Et Statistiques*, *57*(4), 2043–2065.

Janati, H., Muzellec, B., Peyré, G., & Cuturi, M. (2020). Entropic Optimal Transport Between Unbalanced Gaussian Measures Has a Closed Form. *Advances in Neural Information Processing Systems*, *33*.

Jordan, R., Kinderlehrer, D., & Otto, F. (1998). The Variational Formulation of the Fokker–Planck Equation. *SIAM Journal on Mathematical Analysis*, *29*(1), 1–17.

Kondratyev, S., Monsaingeon, L., & Vorotnikov, D. (2016). A New Optimal Transport Distance on the Space of Finite Radon Measures. *Advances in Differential Equations*, *21*(11/12), 1117–1164.

Lavenant, H., Zhang, S., Kim, Y.-H., & Schiebinger, G. (2021). Towards a Mathematical Theory of Trajectory Inference. *Arxiv Preprint Arxiv:2102.09204*.

Liero, M., Mielke, A., & Savaré, G. (2018). Optimal Entropy-Transport Problems and a New Hellinger–Kantorovich Distance Between Positive Measures. *Inventiones Mathematicae*, *211*(3), 969–1117.

Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., & Le, M. (2023). Flow Matching for Generative Modeling. *International Conference on Learning Representations*.

Lu, Y., Lu, J., & Nolen, J. (2019). Accelerating Langevin Sampling with Birth-Death. *Arxiv Preprint Arxiv:1905.09863*.

Léonard, C. (2012). From the Schrödinger Problem to the Monge–Kantorovich Problem. *Journal of Functional Analysis*, *262*(4), 1879–1920.

Léonard, C. (2014). A Survey of the Schrödinger Problem and Some of its Connections with Optimal Transport. *Discrete and Continuous Dynamical Systems – Series a*, *34*(4), 1533–1574.

McCann, R. J. (1997). A Convexity Principle for Interacting Gases. *Advances in Mathematics*, *128*(1), 153–179.

Mei, S., Montanari, A., & Nguyen, P.-M. (2018). A Mean Field View of the Landscape of Two-Layer Neural Networks. *Proceedings of the National Academy of Sciences*, *115*(33), E7665–E7671.

Mena, G., & Niles-Weed, J. (2019). Statistical Bounds for Entropic Optimal Transport: Sample Complexity and the Central Limit Theorem. *Advances in Neural Information Processing Systems*, *32*.

Mikami, T. (2004). Monge's Problem with a Quadratic Cost by the Zero-Noise Limit of h-Path Processes. *Probability Theory and Related Fields*, *129*(2), 245–260.

Nitanda, A., Wu, D., & Suzuki, T. (2022). Convex Analysis of the Mean Field Langevin Dynamics. *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*.

Nutz, M. (2021,). *Introduction to Entropic Optimal Transport*.

Otto, F. (2001). The Geometry of Dissipative Evolution Equations: the Porous Medium Equation. *Communications in Partial Differential Equations*, *26*(1–2), 101–174.

Otto, F., & Villani, C. (2000). Generalization of an Inequality by Talagrand and Links with the Logarithmic Sobolev Inequality. *Journal of Functional Analysis*, *173*(2), 361–400.

Peyré, G., & Cuturi, M. (2019). Computational Optimal Transport. *Foundations and Trends in Machine Learning*, *11*(5–6), 355–607.

Pooladian, A.-A., & Niles-Weed, J. (2021). Entropic Estimation of Optimal Transport Maps. *Arxiv Preprint Arxiv:2109.12004*.

Rotskoff, G., Jelassi, S., Bruna, J., & Vanden-Eijnden, E. (2019). Global Convergence of Neuron Birth-Death Dynamics. *Proceedings of the 36th International Conference on Machine Learning*.

Santambrogio, F. (2015). *Optimal Transport for Applied Mathematicians: 87* (Vol. 87). Birkhäuser.

Schiebinger, G., Shu, J., Tabaka, M., Cleary, B., Subramanian, V., Solomon, A., Gould, J., Liu, S., Lin, S., Berube, P., & others. (2019). Optimal-Transport Analysis of Single-Cell Gene Expression Identifies Developmental Trajectories in Reprogramming. *Cell*, *176*(4), 928–943.

Schrödinger, E. (1932). Sur la théorie relativiste de l'électron et l'interprétation de la mécanique quantique. *Annales De L'institut Henri Poincaré*, *2*(4), 269–310.

Shi, Y., De Bortoli, V., Campbell, A., & Doucet, A. (2023). Diffusion Schrödinger Bridge Matching. *Advances in Neural Information Processing Systems*, *36*.

Sinkhorn, R. (1964). A Relationship Between Arbitrary Positive Matrices and Doubly Stochastic Matrices. *The Annals of Mathematical Statistics*, *35*(2), 876–879.

Tong, A., Fatras, K., Malkin, N., Huguet, G., Zhang, Y., Rector-Brooks, J., Wolf, G., & Bengio, Y. (2024). Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport. *Transactions on Machine Learning Research*.

Tong, A., Huang, J., Wolf, G., Dijk, D. van, & Krishnaswamy, S. (2020). TrajectoryNet: A Dynamic Optimal Transport Network for Modeling Cellular Dynamics. *Proceedings of the 37th International Conference on Machine Learning*, 9526–9536.

Villani, C. (2009). *Optimal Transport: Old and New: 338* (Vol. 338). Springer.

Waddington, C. H. (1957). *The Strategy of the Genes*. George Allen & Unwin.

Weinreb, C., Wolock, S., Tusi, B. K., Socolovsky, M., & Klein, A. M. (2018). Fundamental Limits on Dynamic Inference from Single-Cell Snapshots. *Proceedings of the National Academy of Sciences*, *115*(10), E2467–E2476.

Yeo, G. H. T., Saksena, S. D., & Gifford, D. K. (2021). Generative Modeling of Single-Cell Time Series with PRESCIENT Enables Prediction of Cell Trajectories with Interventions. *Nature Communications*, *12*, 3222.